GEK-83807

DRIVE SYSTEM INSTRUCTIONS

POWER SYSTEM STABILIZER 3S7932LA300G4

NOTE: Includes torsional filter, monitoring circuit, and continuous wave frequency transducer

DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

GENERAL ELECTRIC

(GEK-83807)

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The information herein does not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation, and maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company, Drive Systems Department, USA.

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INTRODUCTION

The Power System Stabilizer (PSS) is designed to introduce a supplementary control signal into continuously acting voltage regulator equipment. The equipment will supplement the conventional voltage regulator control with a control function derived so that, under deviations of generator speed or load, excitation becomes controlled as a composite function of voltage and frequency. Under certain conditions, the composite function has been shown to be an aid to power system stability.

The PSS receives a three-phase signal from the generator potential transformers (PTs). This signal, which is frequency-modulated, is passed through the continuous wave frequency transducer which then has an output frequency equal to the modulation frequency and an amplitude proportional to the deviation in frequency. This signal is then passed through two lead-lag stages in order to obtain the proper phase shift characteristic, a band reject filter to attenuate signals at torsional frequencies, a washout stage to eliminate the effect of component drift and slow system frequency variations, a gain stage to obtain maximum control action and an output stage which interfaces the PSS with continuously acting voltage regulating equipment.

The Power System Stabilizer includes the functions of input transducer, signal conditioning, limiting output circuitry, attenuation of torsional frequencies and torsional protective circuitry. This circuit trips the PSS whenever excessive amplification of torsional frequencies by PSS is detected, or whenever the monitoring circuit fails. Figure 5 is a block diagram of the system, while Figure 8 shows a typical functional elementary diagram.

The Power System Stabilizer will apply the following transfer function to modify the output of the frequency transducer for insertion into the voltage regulator:

$$G(s) = \frac{K\,(T_1S)\,(1 + T_2S)\,(1 + T_4S)}{(1 + T_1S)\,(1 + T_3S)\,(1 + T_5S)}$$

Adjustments are provided to vary the parameters of the PSS so as to permit compensation for variations in the characteristics of the regulator, excitation system, generator and power system to which it is applied. These adjustments can be made by calibrated dial settings which will be accurate to within 5% of the desired parameter value. The adjustments are capable of modifying the transfer function parameters through the following typical ranges:

K = 1 to 100
T_1 = 0.1 to 50 seconds
T_2 = 0.2 to 1.5 seconds
T_3 = 0.02 to 0.1 seconds
T_4 = 0.2 to 1.5 seconds
T_5 = 0.02 to 0.1 seconds

RECEIVING AND HANDLING

The Power System Stabilizer is normally supplied mounted in either the regulator or exciter cubicle. The instruction manual for the particular excitation system used should be consulted for receiving and handling instructions. If the PSS is shipped separately (as in the case of a retrofit system), then immediately upon receipt, the equipment should be carefully unpacked to avoid damage. It should be examined for any damage that might have been sustained in transit. If injury or rough handling is evident, a damage claim should be filed immediately with the transportation company and the nearest General Electric sales office should be notified promptly.

STORAGE

If the equipment is not to be used as soon as it is unpacked, it should be stored in a clean, dry place and protected from accidental damage. Particular care should be exercised to avoid storing the equipment in locations where construction work is in progress. If equipment is to be stored for more than three months, it should be kept heated to at least 5°C above ambient temperature to prevent condensation.

ABBREVIATIONS

The following abbreviations are used in this instruction manual:

AC	Alternating Current
amp	Amperes
db	Decibels
DC	Direct Current
FET	Field Effect Transistor
Hz	Hertz
IC	Integrated Circuit
LED	Light Emitting Diode
MA	Milliampere
MV	Millivolt
PP	Peak-to-Peak
PT	Potential Transformer
PSS	Power System Stabilizer
SCR	Silicon Controlled Rectifier
VAC	Alternating Current Voltage
VCO	Voltage-Controlled Oscillator
VDC	Direct Current Voltage

DESCRIPTION (COMPONENTS)

The Power System Stabilizer consists of:

3S7932LA300G4 Rack Assembly

The Power System Stabilizer is normally mounted in the regulator cubicle and contains the following:

NUMBER	NAME	QUANTITY
304A6033	Continuous Wave Frequency Transducer	1
44C331857-G01	Signal Conditioning Board	2
44C372640-G03	Band Reject Filter	1
44C372640-G04	Band Pass Filter	1
44C331859-G01	Washout and Output Board	1
44C331863-G02	Relay Board	1
44C331861-G01	Ceiling Sensing Board	1
44C372641-G01	PSS Comparator	1

INSTALLATION

Location and Connections

The Power System Stabilizer is normally furnished as an integral part of the excitation and regulation equipment. Refer to the drawings furnished with the excitation system and the instruction manual for the excitation system for location and connection instructions.

Initial Checks

1. Wire check all external connections to the Power System Stabilizer. Be sure that wire size and type are adequate for the expected signal levels. Refer to the connection diagram furnished with the excitation equipment.

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DO NOT HI-POT OR MEGGER ANY CIRCUIT, AS THIS MAY RESULT IN DAMAGE TO COMPONENTS. ANY WIRING OR CIRCUITS TO BE HI-POTTED OR MEGGERED MUST FIRST BE DISCONNECTED FROM THE PSS.

2. Insure that all printed circuit boards are fully inserted.

OPERATION

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MISOPERATION OF THE POWER SYSTEM STABILIZER WHILE CONNECTED TO THE EXCITATION SYSTEM MAY RESULT IN DAMAGE TO THE GENERATOR. DEVIATIONS FROM THE FOLLOWING OPERATIONAL CHECKS AND PROCEDURES SHOULD BE CONSIDERED IN LIGHT OF THIS AND SHOULD BE DISCUSSED WITH THE MANUFACTURER PRIOR TO INITIATION.

Initial Operation

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THE POWER SYSTEM STABILIZER MUST BE OUT OF SERVICE OR THE REGULATOR IN THE DC MODE WHEN CONDUCTING ALL TESTS THRU PARAGRAPH 10. FAILURE TO DO THIS MAY RESULT IN DAMAGE TO THE GENERATOR.

1. Continuous Wave Frequency Transducer Input Check: With the turbine generator at rated speed, close switch L1SW to check input to Continuous Wave Frequency Transducer board (LOC 10501). This can be done by checking the voltages from common to test points TP2, TP3, and TP4 on that board. These are marked "A", "B", and "C", respectively. These voltages should be 2.1 ± 0.2 volts, RMS. They should be 120° apart, with "A" leading "B", and lagging "C". Phase sequence must be correct for proper operation.

   Open switch L1SW and pull the Continuous Wave Frequency Transducer board (LOC 10501).

2. Power Supply Tests: If the PSS does not use the cubicle power supplies, it is necessary to check the built-in power supplies (this does not apply to the Generrex* Excitation System; check elementary diagrams on other excitation systems.)

   Turn on ±15 VDC power using the toggle switch mounted on the faceplate of the Power Supply board.

   Check that the power supply voltages are correct. Adjustment of the ±15 volt supply may be made with board mounted rheostats (refer to Theory of Operation section). The current limit and the crow bar settings are factory set and need no further adjustment.

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If, as on the Generrex Excitation System, the cubicle power supplies are used, ensure that the power supply busses are energized and at the proper level. If adjustment is needed, refer to the excitation system instruction book.

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ON SOME EXCITATION EQUIPMENT, PULLING THE RELAY BOARD INTRODUCES A STEP INTO THE REGULATOR. THIS CAN RESULT IN DAMAGE TO THE GENERATOR. INSURE THAT THE REGULATOR IS IN THE DC MODE OF OPERATION BEFORE PULLING THE RELAY BOARD.

3. Relay Board Tests: If the generator is in the AC mode of operation, transfer to DC. Pull the Relay board (LOC 10512). The Relay board is designed to interface with many excitation systems. Board-mounted shorting clips and rheostats are used to adapt the Relay board to the particular excitation system. Refer to the drawings furnished with the equipment for the settings of F1S, F2S, F3S, F4S, F1P and F2P. Check settings and readjust if necessary. Insert Relay board. Pull Washout and Output board (LOC 10508). Put shorting clip C5S on terminals 1, 2, "not gnd". Insert Extender board in LOC 10508.

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THE REGULATOR MUST BE IN THE DC MODE OF OPERATION FOR THIS TEST. FAILURE TO PLACE THE REGULATOR IN THE DC MODE MAY RESULT IN DAMAGE TO THE GENERATOR.

Insert a signal from pins 11, 12, to pins 3, 4 of the Extender board. This signal can be a DC signal or an oscillating signal (peak value should be less than 10 volts). With the PSS off (as evidenced by the "OFF" LED on the Relay board being lighted) no voltage should exist at the test jack labeled "Out to Regulator" mounted on the Relay board. Pressing the "ON" pushbutton should cause the voltage to be present but attenuated for insertion into the appropriate regulator. If the "ON" LED will not stay energized, press the "Reset" pushbuttons mounted on the faceplate of the Ceiling Sensing board and the faceplate of the PSS Comparator board. If the "ON" pushbutton will still not stay energized, check that the permissive contact is closed (usually 52G line breaker). If the unit is off line, this permissive contact must be jumpered in order to establish that the output relay is working properly. Press the "OFF" pushbutton. The voltage at the test jack should be zero and the "OFF" LED should be lighted. Remove Extender board. Insert Washout and Output board in LOC 10508. On Ceiling Sensing board (LOC 10510), put the delay dial on zero, the positive limit dial on 5 and the negative limit dial on 5. Press the reset pushbutton on the Ceiling Sensing board. Adjust the gain for 1.0 by turning the dial marked "Gain K" on the Washout and Output board to zero. Connect a variable source of DC voltage to "IN1" of the Washout and Output board (C5S on 1, 2) and set the voltage to some value less than 5 volts. Press the "ON" pushbutton on the Relay board. A voltage should exist at the test jack on the Relay board labeled "Out to Regulator". Increase the voltage at "IN1" of the Washout and Output board to 6 volts. After approximately 2.5 seconds, the FAR relay should drop out causing the "OFF" LED to be lighted. Try to reinsert the PSS by pressing the "ON" pushbutton. The PSS should not latch in the "on" position. Remove the voltage from IN1. Press the reset pushbutton on the Ceiling Sensing board. Put the PSS in service by pressing the "ON" pushbutton. The PSS should latch in the "on" position and the "ON" LED should be lighted. Remove the PSS from service by pressing the "OFF" pushbutton. The "OFF" LED should be lighted. Customer controls for the PSS are normally mounted in the control room. Check that these controls can be used to insert and remove the PSS. Leave the PSS in the "off" position.

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THE FOLLOWING GAIN TEST MUST BE PERFORMED WITH THE PSS IN SERVICE AND THE REGULATOR IN THE AC MODE. INSURE THAT THE SHORTING CLIPS AND RHEOSTAT SETTINGS ON THE RELAY BOARD ARE CORRECT FOR THIS EQUIPMENT. INSURE THAT THE CONTINUOUS WAVE FREQUENCY TRANSDUCER IS PULLED. FAILURE TO DO THIS MAY RESULT IN DAMAGE TO THE GENERATOR.

The Relay board contains the necessary interface circuitry so that a one volt change in PSS output voltage will cause a one percent change in generator terminal voltage. With the PSS off, a small step of voltage ($\approx$1 VDC) should be inserted into IN1 of the Washout and Output board. The regulator should be in the automatic mode of operation with the generator preferably off-line. The test may be satisfactorily performed on-line if the reactive current compensator (if any) is bypassed or switched to zero and if the voltage regulator of any other generator units paralleled at terminal voltage are temporarily turned to DC control. Place the PSS in service. Measure the change in the generator terminal voltage. The gain is calculated by dividing the percent change in generator terminal voltage by the step in voltage applied to IN1. The gain should be (-) 1%/volt $\pm$0.1. The minus sign (terminal voltage decreases for a positive input voltage) arises because the signal is inserted into IN1 which bypasses the washout stage. F1P (on the Relay board) can be used to adjust the gain, if necessary. Remove the voltage at IN1.

An attempt has been made to interface with the voltage regulator in such a way that placing the PSS into and out of service will not cause a bump. On some equipments, F2P on the Relay board has been furnished for this purpose. With the voltage regulator in DC (or test if a test position is available on the control switch) pull the Continuous Wave Frequency Transducer (LOC 10501). Put the PSS in service using the pushbuttons on the Relay board. Note whether the AC regulator output voltmeter or transfer voltmeter changes when the PSS is put in service. Use F2P on the Relay board to minimize the change in AC regulator output voltage when the PSS is inserted.

Remove the PSS from service.

4. Continuous Wave Frequency Transducer Board Tests: Check the operation of the Continuous Wave Frequency Transducer board (LOC. 10501). Ascertain that switches R204, R205 and R206 on the R terminal boards are open.

Connect a 3-phase 115 VRMS, 60 Hz signal to terminal points R204B, R205B, and R206B. Phase sequence must be in that order. Insert the Continuous Wave Frequency Transducer board. The output, measured at TP5, should be zero. This test point is mounted on the faceplate of the C. W. F. transducer board and is marked OUT. If necessary, reduce this voltage to zero using potentiometer R2, marked NULL, on the C. W. F. transducer board. Remove the 3-phase 115 VAC source.

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THE PSS MUST BE OUT OF SERVICE FOR THE FREQUENCY RESPONSE TESTS. FAILURE TO DO THIS MAY RESULT IN DAMAGE TO THE GENERATOR.

5. Frequency Response of Main Channel: Record pot settings and jumper positions prior to testing so that control can be reset.

   Set both Signal Conditioning boards (LOC 10502, 03) for lead time constants of 0.26 seconds and lag time constants of 0.03 seconds. This can be accomplished by the following adjustments:

   TCT2V = 0 turns

   TCT2F = 0.9 turns

   TCT3 = 1.5 turns

   TCT4 = 0.9 turns

   TCT5 = 1.5 turns

BIS jumper should be on 1-2 (GRD).

On the Washout and Output board (LOC 10508), set the washout time constant for 2 seconds and the gain for 20. This can be accomplished by placing the TCT1 dial on 6.8 and the gain control at 3.8.

Jumpers should be as follows:

C1S 2-3
C2S 2-3 (150K)
C3S 1-2 (1-50)
C4S 1-2 (1-50)
C5S 2-3 (GRD)

Check the Band Reject Filter (LOC. 10504) to confirm that the shorting clips on 1TB, 2TB, 3TB and 4TB are on terminals 1, 2. The shorting clip on 5TB should be on terminals 2, 3. Pull the Continuous Wave Frequency Transducer board (LOC 10501). Insert the Extender board in place of the Transducer board. Connect an audio oscillator to terminal 11, 12 and ground terminal 3 or 4 of the Extender board. Measure the frequency response of the PSS main channel from 0.16 Hz to 143 Hz measuring oscillator output and the output of the Washout and Output board. The latter can be measured on the "Out 1" test jack of the Washout and Output board. The oscillator output level should be held to those shown in Figure 1 to avoid saturation of the amplifiers. Record data and compare gain to values shown in Figure 1.

Input and output signals should be observed with an oscilloscope to be sure that saturation or distortion is not present.

FREQ HZ	V(IN) VOLTS (ZERO TO PEAK)	V(OUT) VOLTS (ZERO TO PEAK)	GAIN	NOMINAL GAIN (1)	
0.1	0.2			14.1	
0.2	0.2			18.2	
0.4	0.2			24.8	
1.0	0.2			58.2	
2.0	0.02			164.1	
3.0	0.02			245.5	
5.0	0.02			197.2	
7.0	0.2			12.3	
10	0.2			33.9	
12	0.02			69.2	
15	0.02			36.7	
17	0.02			46.8	
20	0.02			51.9	
25	0.02			82.2	
30	0.02			95.5	
35	0.02			83.2	
40	0.02			55.0	
45	0.2			32.4	
50	0.2			14.6	
55	0.2			5.2	
65	0.2			3.6	
70	0.2			4.3	
80	0.2			5.8	
90	0.2			7.4	
100	0.2			9.2	

T_1 = 2 SEC. T2 = T4 = 0.26 SEC
T_3 = T5 = 0.03 SEC.
GAIN = 20

Gain is measured from the output of the Continuous Wave Frequency Transducer to the output of the Washout and Output board. Use an oscilloscope to ascertain that nothing is saturating.

Figure 1
Frequency Response of Main Channel

6. Frequency Response of Monitor Channel: Using the same settings as 5, make a frequency response check of the monitor (or protective) channel. Set the gain pots (11P and 31P on the PSS Comparator (LOC 10506) to minimum (fully CCW). The signals of concern are the voltages at 2TP and 3TP of the PSS Comparator board.

   Use an oscilloscope to make sure saturation or distortion is not present.

   Set the audio oscillator frequency for 15 Hz. Balance the voltages at 2TP and 3TP on the PSS Comparator board by increasing the lesser of the two voltages (use either 11P or 31P). Using Figure 2 as a guide, determine the amplitude response of 2TP and 3TP as a function of frequency. The amplitude of the voltage at 2TP and 3TP should not differ by more than 0.25 volt over the frequency range shown in Figure 2. The gain at 2TP should be calculated and compared to the nominal gain shown in Figure 2. The actual gain and nominal gain should not differ by more than 2 to 1. The nominal gain as a function of frequency is shown in Figure 3 for the main channel and the protective channel.

7. PSS Comparator Tests: Check out the trip circuit of the PSS Comparator. The trip reference voltage at 4TP is five times the voltage difference in 2TP and 3TP at which a trip will occur. Set the trip reference adjustment, 51P, for 2.5 volts at 4TP. With the oscillator at 15 Hz, slowly adjust (either increase or decrease) TCT2F on either Signal Conditioning board until the voltage at 5TP of the PSS Comparator begins to rise. The voltage at 2TP and 3TP should differ by 0.4 to 0.6 volt. The trip light should begin to flash as 5TP goes positive. Return TCT2F to its original position and reset the trip.

   Final settings of the two Signal Conditioning boards should be identical. The final setting of the PSS Comparator trip reference, 51P, should be coordinated with the infinite shaft life torque levels of the machine. Refer to the manufacturer for this setting. Remove the audio oscillator.

FREQ. HERTZ	V(IN) PEAK MILLIVOLTS	2TP -MILLIVOLTS	3TP +MILLIVOLTS	DIFF |2TP|-|3TP|	GAIN V (2TP)/V(IN)	NOMINAL GAIN
1	3.0					0.3
2	1.0					1.8
4	0.2					8.5
8	0.2					23.2
10	0.2					27.9
12	0.2					28.0
15	0.2					27.5
18	0.2					26.6
20	0.2					24.3
25	0.2					18.2
30	0.2					11.1
40	0.2					4.0
45	2.0					2.0

NOTE: See instructions for gains and time constants used for nominal gain figures.

2TP-3TP difference should not exceed 0.25 volts at any frequency.

Figure 2
Frequency Response At Comparator

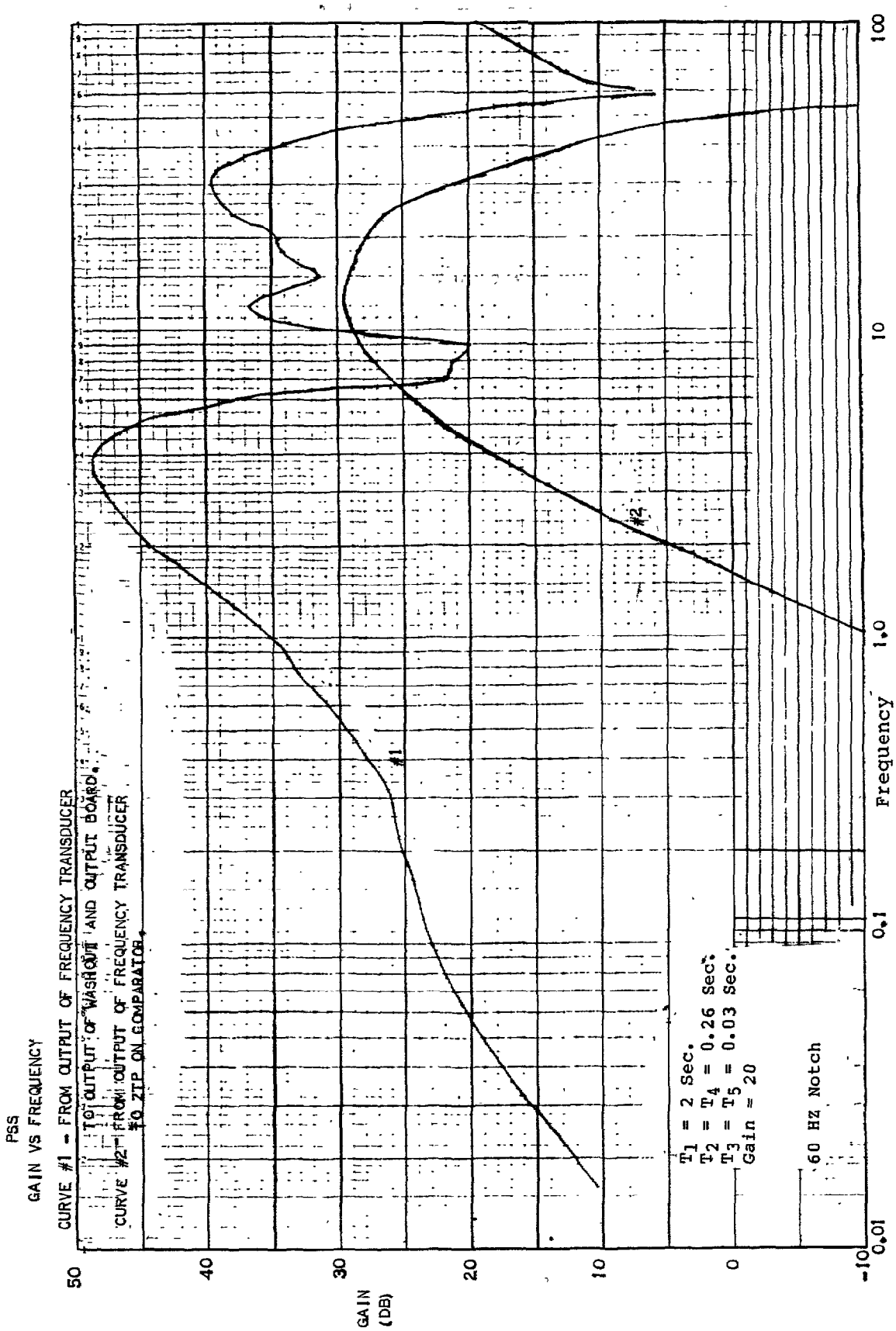


Figure 3
Frequency (Hz)

8. Ceiling Sensing Tests: Check out the Ceiling Sensing board. Pull the Ceiling Sensing board and verify that the shorting clip on E1S is between 1 and 2. Reinsert the board. Set both the positive limit (E2P) and negative limit (E3P) for 500 (5.0 turns). Set the time delay on the Ceiling Sensing board for minimum (0). Insert a variable DC voltage into IN1 of the Washout and Output board. Place the regulator in the manual mode of operation.

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THE REGULATOR MUST BE IN THE DC MODE OF OPERATION FOR THIS TEST. FAILURE TO DO THIS CAN RESULT IN DAMAGE TO THE GENERATOR.

Place the PSS in service by pressing the "ON" pushbutton located on the Relay board. Slowly increase the voltage until the PSS trips out of service as evidenced by the "OFF" pushbutton being lighted. The voltage at OUT1 on the Washout and Output board should be 5 volts ± 0.25 volts. Remove the voltage. Pressing the "ON" pushbutton on the Relay board should have no effect. Press the reset pushbutton on the Ceiling Sensing board. Pressing the "ON" pushbutton on the Relay board should now cause the PSS to be placed in service. Reverse the voltage into IN1 of the Washout and Output board. The same procedure can be followed to verify that this part of the circuit is working. Adjust the time delay (E1P) on the Ceiling Sensing board for 300 (3 turns). Remove the voltage at IN1 of the Washout and Output board and press the reset pushbutton on the Ceiling Sensing board. Place the PSS in service (regulator in manual). Put a voltage into IN1 of sufficient magnitude to cause a trip. The PSS should trip to "OFF" approximately 10 seconds after insertion of the signal. Reset the Ceiling Sensing board by pressing the reset pushbutton on the faceplate. Place the positive and negative limit dials on 1000 (10 turns) and the time delay dial on zero. Final settings can only be made after on-line test and/or system studies.

9. Washout and Output Limit Test: On Washout and Output board, set negative limit dial and positive limit dial on 500 (5 turns). Insert a variable DC voltage into IN1. Monitor the output voltage at OUT1 using an oscilloscope or voltmeter. Increase the DC voltage slowly and note the voltage level at which the output fails to follow the input voltage (output voltage is limited). This voltage should be 4.75 volts ± 0.5 volts. Reverse the voltage and perform the same test. The output voltage should be of the opposite polarity but same magnitude (within ± 0.5 volts. Remove the voltage and Extender board. Reinsert the Continuous Wave Frequency Transducer board. Place the regulator in the AC mode of operation. Insure that the PSS is off. Set the positive and negative limit dials on 1000 (10 turns) and the time delay dial on zero. Final settings can only be made after on-line tests and/or system studies.